

Environmental Health Monitoring App

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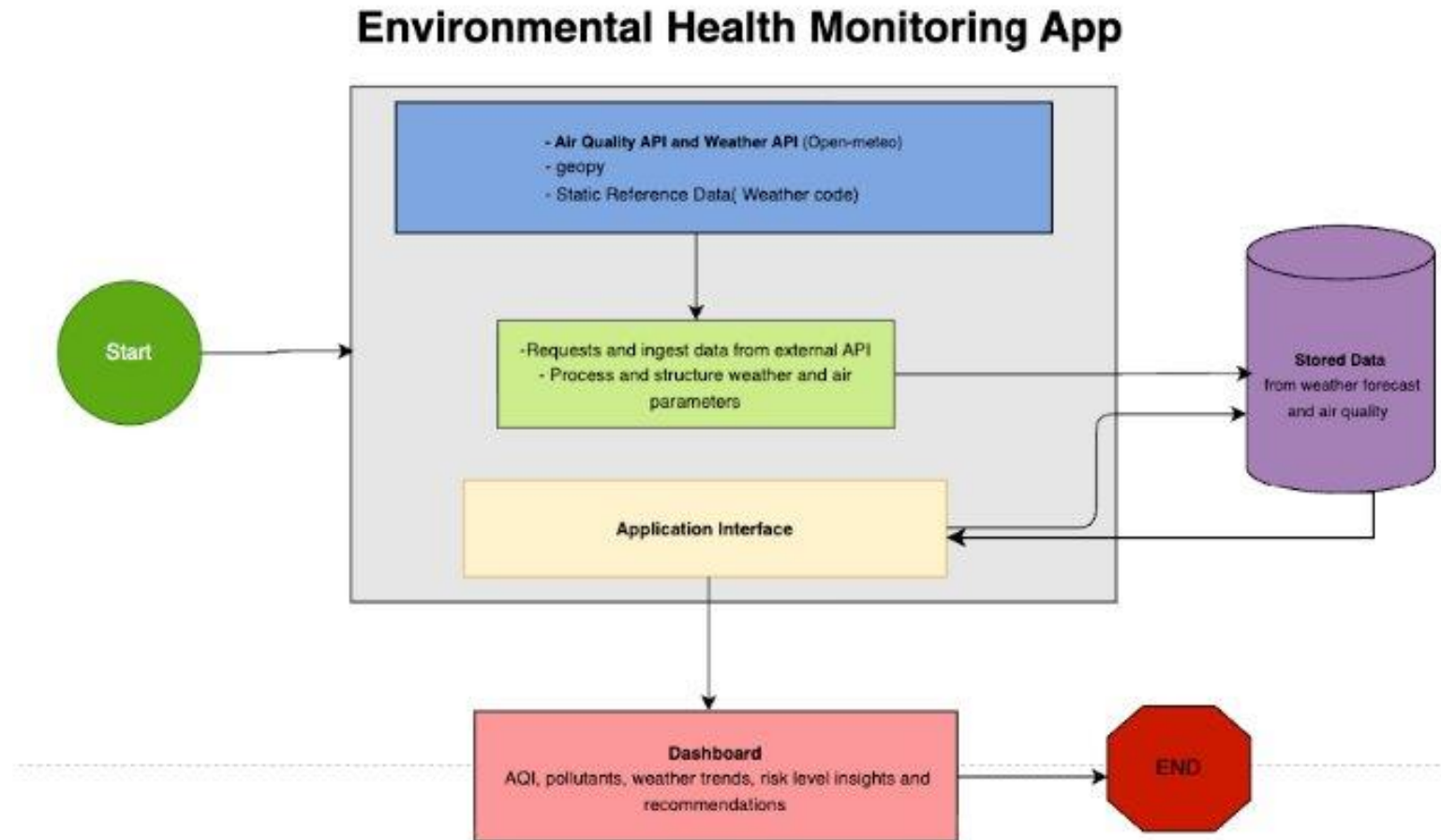
Business Problem

Air quality and weather conditions directly affect human health, productivity, and quality of life. Despite the public availability of environmental data from various sources, that data is fragmented, raw, and inaccessible to non-technical users. People living in areas prone to heatwaves, wildfire smoke, or high particulate pollution have no easy way to understand what the numbers mean for them personally, let alone act on them in real time.

Objectives

- Develop an environmental health dashboard for users
- Integrate real-time and historical air quality data using the Open-Meteo Air quality API and geopy library
- Integrate weather data (temperature, wind, humidity, precipitation) using Open-Meteo weather API
- Analyze relationship between weather conditions and air pollution levels
- Provide health-focused environmental risk indicators for users

Technical Architecture Diagram



 weather_observations	
  id	int4
 location	text
 fetched_at	timestampz
 temperature_f	numeric
 feels_like_f	numeric
 weather_code	int4
 condition	text
 emoji	text
 is_day	bool
 wind_speed_mph	numeric
 humidity_pct	numeric
 visibility_mi	numeric
 pressure_inhg	numeric
 dew_point_f	numeric
 timezone	text
 daily_forecast	jsonb
 created_at	timestampz

 air_quality_observations	
  id	int4
 location	text
 fetched_at	timestampz
 us_aqi	int4
 aqi_category	text
 aqi_color	text
 pm2_5	numeric
 pm10	numeric
 ozone_ppb	numeric
 no2_ppb	numeric
 so2_ppb	numeric
 co_ppb	numeric
 primary_pollutant	text
 primary_value	numeric
 hourly_trend	jsonb
 created_at	timestampz

[git repository link](#)

develop

2 Branches

0 Tags

Go to file

T

Add file

Code



laxmiadh08

updated proposal document

fa6ade5 · 10 minutes ago

🕒 20 Commits

 Diagram	updated proposal document	10 minutes ago
 data	all new files updated	last week
 docs	updated proposal document	10 minutes ago
 images	added architecture diagram	1 hour ago
 .gitignore	all new files updated	last week
 README.md	added architecture diagram	1 hour ago
 app.py	Merge branch 'feature/load-script' into develop	5 days ago
 load.py	load script update	5 days ago
 main.py	readme content added	5 days ago
 requirements.txt	all new files updated	last week

Environmental Health Monitor

Location Search 

Today

This Week

 **Louisville, Jefferson County, Kentucky** - Monday, June 08 2026 09:45 PM

Current Weather
75°F  **Mainly clear, partly cloudy, and overcast**
Feels like 80 °F

Air Quality 31 AQ	Wind 7 mph	Humidity 86 %	Visibility 9.9 mi	Pressure 29.52 in	Dew point 70.6 °F
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 **Low Risk**
Temp: 75°F | Humidity: 86% | AQI: 31

 **Primary Pollutant: CO 149.0 µg/m³ · AQI: 31**

Conditions are generally safe for outdoor activities. Continue to stay hydrated and monitor conditions if planning extended exertion.

 Air quality and temperature are within safe ranges.

 Maintain normal hydration habits.

 Outdoor exercise is suitable for all groups.

 Check back later for any changing conditions.

Current Pollutant Levels

O ₃ 58.0 ppb	PM2.5 4.2 µg/m³	PM10 4.6 µg/m³	NO ₂ 3.6 ppb	CO 149.0 ppb	SO ₂ 0.5 ppb
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Hourly Air Quality Index



Environmental Health Monitor

Location Search 

Today

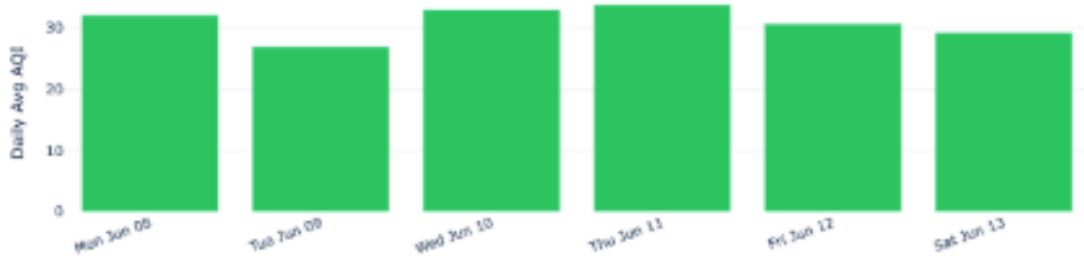
This Week

 **Louisville, Jefferson County, Kentucky** - Monday, June 08 2026 09:45 PM

7-Day Forecast

Mon	Tue	Wed	Thu	Fri	Sat	Sun
 81° 73°	 83° 71°	 89° 70°	 92° 74°	 88° 72°	 87° 66°	 80° 71°

7-Day Average Air Quality Index



Environmental Health Monitor

Location

Delhi

Search



Louisville

Today

This Week

Environmental Health Monitor

Location

Delhi

Search

Louisville

Today

This Week

New Delhi, Delhi, India · Tuesday, June 09 2026

07:15 AM

Current Weather

91°F Clear sky
Feels like 92°F

Air Quality

524 AQI

Wind

6 mph

Humidity

37%

Visibility

7.9 mi

Pressure

28.81 in

Dew point

61°F



Extreme Danger

Temp: 91°F | Humidity: 37% | AQI: 524



Primary Pollutant: PM10 635.2 µg/m³ · AQI: 524

Total ban on outdoor labor and exercise. Standard fans are insufficient — air-conditioned spaces with air purifiers are required. Prolonged outdoor exposure risks heat stroke and severe respiratory distress.

- Do NOT work or exercise outdoors under any circumstances.
- Stay in AC spaces; use air purifiers rated for PM2.5/PM10.
- Drink at least 1 litre of water per hour if movement is unavoidable.
- Check on elderly, children, and those with heart or lung conditions every hour.
- N95/KN95 masks are mandatory for any unavoidable outdoor exposure.

Current Pollutant Levels

O₃

28.0 ppb

PM2.5

149.0 µg/m³

PM10

635.2 µg/m³

NO₂

62.5 ppb

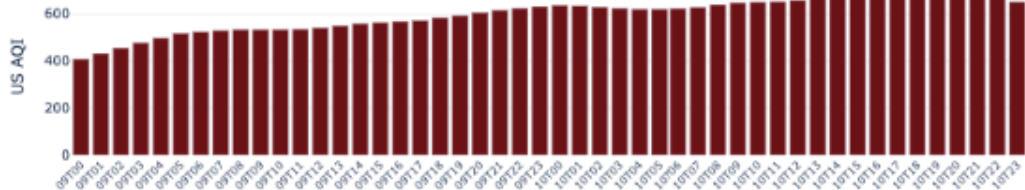
CO

504.0 ppb

SO₂

28.1 ppb

Hourly Air Quality Index



Good Moderate Unhealthy for Sensitive Groups Unhealthy Very Unhealthy Hazardous

Location

Lahore

Search

Louisville

Today

This Week

لاہور، تحصیل لاہور شہر، ضلع لاہور · Monday, June 15 2026

02:

Current Weather

80°F Clear sky
Feels like 87°F

Air Quality

108 AQI

Wind

2 mph

Humidity

69%

Visibility

6.1 mi

Pressure

28.82 in

Dew point

69.1°F



High Risk

Temp: 80°F | Humidity: 69% | AQI: 108



Primary Pollutant: CO 852.0 µg/m³ · AQI: 108

Vulnerable groups — elderly, children, pregnant women, and those with respiratory or cardiovascular conditions — must stay indoors in air-filtered, cooled spaces.

- Limit all outdoor activity to before 8 AM or after sunset.
- Sensitive groups should remain indoors with windows closed.
- Run HVAC on recirculation; avoid bringing outdoor air inside.
- Stay well-hydrated; avoid caffeine and alcohol.
- Wear a mask (N95/KN95) if outdoor exposure is unavoidable.

Current Pollutant Levels

O₃

25.0 ppb

PM2.5

63.8 µg/m³

PM10

75.9 µg/m³

NO₂

78.8 ppb

CO

852.0 ppb

SO₂

18.2 ppb

Hourly Air Quality Index

